

API SECURITY RISK ANALYSIS REPORT

TARGET API: JSONPLACEHOLDER.TYPICODE.COM



Assessed By: Xhanti



CIN: FIT/APR26/CS8004



Date: 26 May 2026



Program: Future Interns
– Cyber Security Task 3



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INTRODUCTION

This report presents a security risk analysis of the JSONPlaceholder public API at jsonplaceholder.typicode.com.

What is API Security?

API (Application Programming Interface) security refers to the protection of APIs from attacks and misuse. Insecure APIs can expose sensitive data and allow unauthorized access.

API Endpoints Tested:

- /users – User personal data
- /posts – User posts data
- /comments – User comments data

Tools Used:

- Browser DevTools (Microsoft Edge)
- JSONPlaceholder Public API
- Canva (report design)

Objective:

To identify API security risks, analyze data exposure, and provide remediation recommendations.

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API ENDPOINTS TESTED

Base URL: <https://jsonplaceholder.typicode.com>

ENDPOINT 1: /users

URL: <https://jsonplaceholder.typicode.com/users>

Method: GET

Authentication Required: **✗** NONE

Data Exposed: Names, emails, phone numbers,
addresses, GPS coordinates, company info

ENDPOINT 2: /posts

URL: <https://jsonplaceholder.typicode.com/posts>

Method: GET

Authentication Required: **✗** NONE

Data Exposed: User IDs, post titles,
post body content

ENDPOINT 3: /comments

URL: <https://jsonplaceholder.typicode.com/comments>

Method: GET

Authentication Required: **✗** NONE

Data Exposed: Names, email addresses,
comment content

**⚠ ALL 3 ENDPOINTS ARE PUBLICLY ACCESSIBLE
WITHOUT ANY AUTHENTICATION OR AUTHORIZATION!**

API SECURITY RISKS IDENTIFIED

● RISK 1: NO AUTHENTICATION (HIGH)

Any user can access all data without logging in or providing credentials.

Impact: Unauthorized data access

● RISK 2: SENSITIVE DATA EXPOSURE (HIGH)

Full names, emails, phone numbers, home addresses and GPS coordinates are freely accessible.

Impact: Privacy violation, identity theft

● RISK 3: GPS LOCATION DATA EXPOSED (HIGH)

Latitude and longitude coordinates visible for all users.

Impact: Physical safety risk

● RISK 4: NO ACCESS CONTROL (HIGH)

Any person can access any user's data.

No role-based access control in place.

Impact: Data breach

● RISK 5: NO RATE LIMITING (MEDIUM)

Unlimited requests can be made to the API with no restrictions.

Impact: DDoS attacks, data harvesting

● RISK 6: NO INPUT VALIDATION (MEDIUM)

API accepts requests without validating or sanitizing input data.

Impact: Injection attacks possible

RISK SUMMARY

Total Risks Identified: 6

● HIGH RISK – 4

- No Authentication on any endpoint
- Sensitive personal data exposed
- GPS location coordinates exposed
- No access control implemented

● MEDIUM RISK – 2

- No rate limiting on API requests
- No input validation on endpoints

RISK SCORE: HIGH

This API poses serious security risks and should not be used in production without implementing proper security controls.

DATA EXPOSURE SUMMARY:

- 10 users personal data exposed
- Names & usernames: ✓ EXPOSED
 - Email addresses: ✓ EXPOSED
 - Phone numbers: ✓ EXPOSED
 - Home addresses: ✓ EXPOSED
 - GPS coordinates: ✓ EXPOSED
 - Company details: ✓ EXPOSED

RECOMMENDATIONS

1. IMPLEMENT AUTHENTICATION (HIGH)

Require API keys or OAuth tokens for all endpoint access.

Fix: Add JWT authentication to all API endpoints.

2. RESTRICT DATA EXPOSURE (HIGH)

Never expose sensitive personal data in public API responses.

Fix: Remove phone, address and GPS data from public endpoints.

3. IMPLEMENT ACCESS CONTROL (HIGH)

Users should only access their own data, not others.

Fix: Add role-based access control (RBAC) to all endpoints.

4. REMOVE GPS DATA (HIGH)

Location data should never be exposed in API responses.

Fix: Remove lat/lng fields from all public responses.

5. ADD RATE LIMITING (MEDIUM)

Limit requests per IP address to prevent abuse.

Fix: Implement max 100 requests per hour per IP address.

6. ADD INPUT VALIDATION (MEDIUM)

Validate and sanitize all input before processing.

Fix: Use input validation library on all API endpoints.

CONCLUSION

This API security risk analysis of jsonplaceholder.typicode.com identified 6 security vulnerabilities across 3 API endpoints.

Key Findings:

- 4 High Risk vulnerabilities found
- 2 Medium Risk vulnerabilities found
- All 3 endpoints lack authentication
- Sensitive personal data freely exposed

Most Critical Issues:

- No authentication on any endpoint
- Full user personal data exposed
- GPS coordinates accessible to anyone

This API should NOT be used in a real production environment without implementing:

- Authentication & Authorization
- Data minimization
- Rate limiting
- Input validation

Immediate security improvements are strongly recommended to protect user privacy and prevent data breaches.

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